

1. List 3 real-world devices in your home or college that use SMPS (Switched Mode Power Supply) and 2 devices that use UPS (Uninterruptible Power Supply), explaining in one line why each device needs that type of power supply.

A) Devices using SMPS (Switched Mode Power Supply)

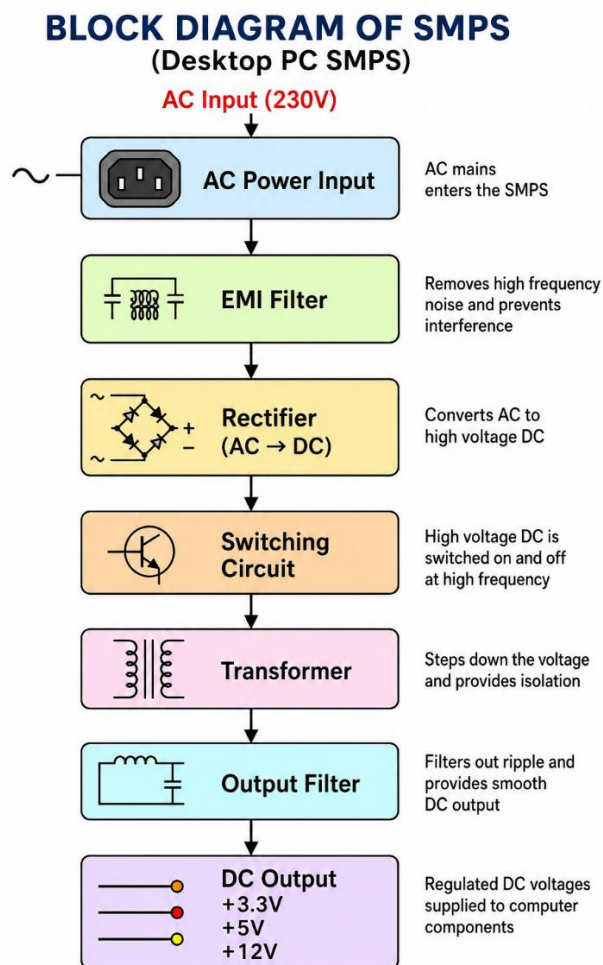
Device	Why it needs SMPS
Desktop Computer	Converts AC electricity into multiple DC voltages required by computer components.
LED Television	Provides efficient and stable power while reducing energy consumption.
Wi-Fi Router	Converts AC power into low-voltage DC power needed by the router.

B) Devices using UPS (Uninterruptible Power Supply)

Device	Why it needs UPS
Desktop Computer	Prevents sudden shutdown and protects data during power cuts.
CCTV Security System	Keeps security cameras running during electricity failures.

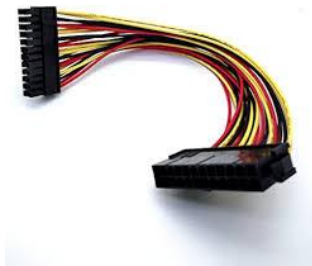
2. Draw and label a diagram showing the main internal components of a typical SMPS unit found in a desktop PC, including AC input, rectifier, transformer, and DC output sections.

Hint: You can hand-draw and upload a photo, or use any free diagram tool like draw.io.



3. Take photos or find clear images online of the following power connectors: ATX 24-pin motherboard connector, SATA power connector, Molex connector, and PCIe 6/8-pin connector. For each, write its name, the devices it powers, and one safety tip for connecting/disconnecting it.

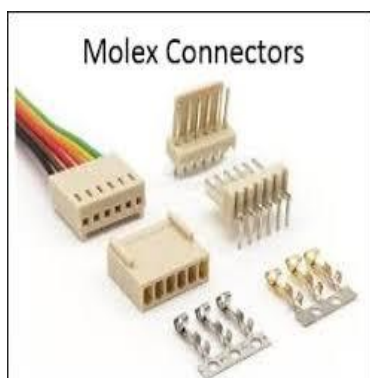
Connector Name	Devices It Powers	Safety Tip
ATX 24-pin Connector	Motherboard	Turn off the power supply before connecting or disconnecting.
SATA Power Connector	SSD, HDD, DVD Drive	Never force the connector into the port.
Molex Connector	Cooling fans and older devices	Pull from the plastic connector, not the wires.
PCIe 6/8-pin Connector	Graphics Card (GPU)	Ensure the connector is fully locked in place.



ATX 24-pin Connector



SATA Power Connector



Molex Connector



PCIe 6/8-pin Connector

4. Imagine you are setting up a new gaming PC. List the steps you would follow to connect the SMPS to the motherboard, graphics card, and storage devices, mentioning which specific cables and connectors are used for each.

Step 1: Install the SMPS

- Place the SMPS inside the computer case.
- Secure it using four screws.

Step 2: Connect the Motherboard

- Use the **24-pin ATX cable**.
- Connect it to the motherboard power socket.

Step 3: Connect CPU Power

- Use the **8-pin EPS (CPU power) cable**.
- Connect it near the CPU socket.

Step 4: Connect the Graphics Card

- Use the **PCIe 6-pin or 8-pin cable**.
- Connect it to the graphics card.

Step 5: Connect Storage Devices

- Use **SATA power cables**.
- Connect them to SSDs and HDDs.

Step 6: Connect Extra Components

- Use Molex connectors if additional fans or accessories require them.

Step 7: Check Connections

- Verify that all cables are securely connected.
- Organize cables properly.

Step 8: Power On the Computer

- Connect the external power cable.
- Turn on the SMPS switch.
- Start the computer and test the system.